

Test sheet

Functions

1. In the binomial expansion of $(2 + x/2)^n$, the first three terms in increasing powers of x are $b + abx + (9/8)abx^2$. Find the values of the constants n , a and b . [8 marks]

2. Hence find and simplify an expression for $g(x)$. [2 marks]

3. Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1} . [5 marks]

4. The functions f and fg are defined by $f(x) = e^{(x^2 + 3)}$ for $x < 0$ and $fg(x) = e^{(2x)}$ for $x > 3/2$. Explain why f^{-1} exists. [1 mark]

5. The diagram shows part of the curve $y = 32x - 4x^2 - 48$ and the line AB . The curve and the line AB meet the x -axis at A and meet again at the point $B(5, 12)$. The line CD extended is parallel to the y -axis and passes through the maximum point of the curve. Find the area of the shaded region. [9 marks]

6. Find the value of x for which $y = 17$ [2 marks]

7. Find the equation of the line in part (a) and hence find the values of A and k. Give each value correct to 1 significant figure. [5 marks]

Answer key

1. $n = 4$, $a = 1$, $b = 2$ (or equivalent values)
2. $g(x) = -\sqrt{2x - 3}$
3. $f^{-1}(x) = -\sqrt{\ln(x) - 3}$, domain $x > e^3$, range $f^{-1} < 0$
4. f is one-to-one (or one-one) on its domain
5. $40/3$ or $13.3[33\dots]$
6. x in range $33 \leq x \leq 37.5$
7. $\ln(y) = 0.03x + 1.8$, $A = 6$ or 7 , $k = 0.03$ or 0.02